

Hooks:

1. Program to write ISR for “DIV BY Zero” exception i.e., INT#0 and hook it to print custom message “Cannot be divided by 0”.
2. **Write an assembly language program to hook the *Overflow Exception* (INT 4).**

The program should modify the Interrupt Vector Table (IVT) so that a **user-defined Interrupt Service Routine (ISR)** is executed whenever an overflow condition occurs.

3. **Write an assembly language program to hook interrupt 0x65 with multiple user-defined services.** The custom Interrupt Service Routine (ISR) must **preserve all registers except AX**.

The interrupt should provide the following services, selected using the value in the **AH register**:

Service Code (AH)	Description
0x01	Add two words pointed to by SI and DI and store the result in the memory location pointed to by BX . $[BX] \leftarrow [SI] + [DI]$
0x02	Multiply two words pointed to by SI and DI and store the result in the memory location pointed to by BX . $[BX] \leftarrow [SI] \times [DI]$
0x03	Divide the word pointed to by SI by the byte pointed to by DI and store the quotient and remainder in registers AL and AH , respectively. $AL \leftarrow [SI] / [DI]$ $AH \leftarrow [SI] \% [DI]$